Table 1: Variable definitions and sources of data used in the study. This table summarizes the variable definitions and sources of data used in our study.

Variables	Definition
Parent Debt Ratio	Ratio of the parent's non-deposit debt liabilities to the BHC's assets, measured quarterly as: (BHCP2309 (parent company's borrowings of commercial paper with a remaining maturity of one year or less) + BHCP2332 (parent company's other borrowings with a remaining maturity of one year or less) + BHCP0368 (parent company's other borrowed money with a remaining maturity of more than one year) + BHCP4062 (parent company's subordinated notes and debentures) + BHCP0279 (parent company's securities sold under agreements to repurchase)) / BHCK2170 (BHC's total assets)
Loan Lease Yield	Ratio of the subsidiary bank's total interest income from loans to its total loans: (RIAD4010 (total interest and fee income on loans)+ RIAD4065) / RCFD2122 (total loans and leases held for investment and held for sale) for FFIEC 031, or (RIAD4010 (total interest and fee income on loans)+ RIAD4065)/ RCON2122 (total loans and leases held for investment and held for sale) for FFIEC 041. Because RIAD4010 and RIAD4065 are reported as year-to-date income, we convert these variables into quarterly amounts before calculating the ratio.
Provision Ratio	Ratio of provisions to total loans: RIAD4230 (provisions for credit losses) / RCFD2122 (total loans and leases held for investment and held for sale) for FFIEC 031, or RIAD4230 (provisions for credit losses) / RCON2122 (total loans and leases held for investment and held for sale) for FFIEC 041. Because RIAD4230 is reported as year-to-date income, we convert this variable into quarterly amount before calculating the ratio.
Leverage	Leverage is the subsidiary bank's debt-to-equity ratio: RCON2948 (bank subsidiary's total liabilities, FFIEC 041) / (RCON2170 (bank subsidiary's total assets, FFIEC 041) − RCON2948 (bank subsidiary's total liabilities, FFIEC 041)), or RCFD2948 (bank subsidiary's total liabilities, FFIEC 031) / (RCFD2170 (bank subsidiary's total assets, FFIEC 031) − RCFD2948 (bank subsidiary's total liabilities, FFIEC 031))
Z-Score	Z-Score is a proxy for bank stability, estimated as return on assets plus the equity-to-asset ratio, scaled by the standard deviation of return on assets. The equity-to-asset ratio is calculated as total equity (RCFD3210, total bank equity capital at the end of the current period) divided by total assets (RCFD2170) for FFIEC 031, or total equity (RCON3210, total bank equity capital at the end of the current period) divided by total assets (RCON2170) for FFIEC 041. To reduce skewness, we take the logarithm of this measure.
Z-Score (CE)	Z-Score (CE) is a proxy for bank stability, estimated as return on assets plus the equity-to-asset ratio, scaled by the standard deviation of return on assets. The equity-to-asset ratio is calculated as book equity (RCFD3632 (retained earnings) + RCFD3839 (surplus, excluding all surplus related to preferred stock) + RCFD3230 (common stock)) divided by total assets (RCFD2170) for FFIEC 031, or book equity (RCON3632 (retained earnings) + RCON3839 (surplus, excluding all surplus related to preferred stock) + RCON3230 (common stock)) divided by total assets (RCON2170) for FFIEC 041. To reduce skewness, we take the logarithm of this measure.
Ln TA	Total assets is a proxy for subsidiary bank size. It is the logarithm of the subsidiary bank's total assets: ln(RCON2170 (bank subsidiary's total assets, FFIEC 041)), or ln(RCFD2170 (bank subsidiary's total assets, FFIEC 031))

HQLA Ratio	HQLA_ratio is the ratio of the BHC's high-quality liquid assets to the BHC's total assets: (BHCM3531 (U.S. Treasury securities) + BHCM3532 (U.S. government agency obligations, excluding mortgage-backed securities) + BHCKG379 (residential pass-through MBS issued or guaranteed by FNMA, FHLMC, or GNMA) + BHCKG380 (other residential MBS issued or guaranteed by U.S. government agencies or sponsored agencies, including CMOs, REMICs, and stripped MBS) + BHCKK197 (commercial MBS issued or guaranteed by U.S. government agencies or sponsored agencies) + BHCKG301 (residential pass-through securities guaranteed by GNMA, held to maturity) + BHCKG303 (residential pass-through securities guaranteed by GNMA, available for sale) + BHCKG305 (residential pass-through securities issued by FNMA and FHLMC, held to maturity) + BHCKG307 (residential pass-through securities issued by FNMA and FHLMC, available for sale) + BHCKKX53 (residential pass-through securities guaranteed by GNMA, issued by FNMA and FHLMC, and other pass-through securities, held to maturity) + BHCKKX55 (residential pass-through securities guaranteed by GNMA, issued by FNMA and FHLMC, and other pass-through securities, available for sale) + BHCKG313 (other residential MBS issued or guaranteed by U.S. government agencies or sponsored agencies, held to maturity) + BHCKG315 (other residential MBS issued or guaranteed by U.S. government agencies or sponsored agencies, available for sale) + BHCKK143 (commercial pass-through securities issued or guaranteed by FNMA, FHLMC, or GNMA, held to maturity) + BHCKK145 (commercial pass-through securities issued or guaranteed by FNMA, FHLMC, or GNMA, available for sale) + BHCKK151 (other commercial MBS issued or guaranteed by U.S. government agencies or sponsored agencies, held to maturity) + BHCKK153 (other commercial MBS issued or guaranteed by U.S. government agencies or sponsored agencies, available for sale) + BHCKHT51 (U.S. government agency and sponsored agency obligations, held to maturity) + BHCKHT53 (U.S. government agency and sponsored agency obligations, available for sale) + RCFD0090 (balances due from Federal Reserve Banks) for FFIEC 031, or RCON0090 (balances due from Federal Reserve Banks) for FFIEC 041 + BHCK0213 (U.S. government agency and sponsored agency obligations, excluding mortgage-backed securities, held to maturity) + BHCK1287 (U.S. government agency and sponsored agency obligations, excluding mortgage-backed securities, available for sale)) / BHCK2170 (BHC's total assets)
TobinQ	Tobin's Q is the group-level consolidated Tobin's Q, defined as the ratio of the market value of the BHC (MV) to the replacement cost of its assets (RC). It is estimated as tobinq_ultp = (AT + ME − BE) / AT, where AT is total assets, ME is the market value of equity at year end (stock price multiplied by common shares outstanding), and BE is book equity (stockholders' equity + deferred taxes + investment tax credit − preferred stock). Calculated using the WRDS code; see link for details: https://wrds-www.wharton.upenn.edu/pages/wrds-research/applications/risk-and-valuation-measures/tobins-q-altman-z-score-and-companys-age/

Table 2

Summary statistics of key characteristics. This table summarizes the sample statistics of key subsidiary bank-level and BHC-level characteristics used in our study. Panel A uses the full sample of subsidiary banks and BHCs during the 2011-2021Q2 period. Panel B uses the reduced sample for difference-indifferences models over the 2012-2013 (pre-deregulation) and 2016-2017 (post-deregulation) periods. Panel C uses the collapsed sample of BHCs in the treatment group for difference-indifferences models over the 2012-2013 (pre-deregulation) and 2016-2017 (post-deregulation) periods. The full sample comprise 298 unique BHCs and 579 subsidiary banks.

Panel A: Full Sample (2011-2021Q2)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
VARIABLES	N	mean	sd	p10	p25	p50	p75	p90	min	max
Loan Lease Yield	8,907	0.0482	0.0157	0.0372	0.0415	0.0459	0.0509	0.0571	0.0184	0.1364
Provision Ratio	8,907	0.0032	0.0068	-0.0011	0.0002	0.0016	0.0037	0.0087	-0.0099	0.0399
Z-Score	9,871	3.6525	3.68889	3.0928	3.3643	3.6045	3.8726	4.2256	2.0747	6.2843
Z-Score (CE)	9,871	3.6545	0.5785	3.0986	3.3667	3.6063	3.8760	4.2357	2.1379	6.1054
Leverage	9,898	7.2881	3.0621	0.7966	6.3252	7.7454	9.0627	10.3153	0.0076	14.2192
Parent Debt Ratio	9,810	0.0187	0.0328	0.0000	0.0000	0.0084	0.0195	0.0543	0.0000	0.3356
TobinQ	9,898	1.0320	0.0540	0.9795	1.0000	1.0230	1.0531	1.0877	0.9122	1.5686
Ln TA	9,898	15.0257	2.2820	12.5406	13.9228	14.9970	16.3062	17.7974	7.2167	21.8888
HQLA Ratio	9,810	0.1542	0.0811	0.0664	0.0943	0.1391	0.1993	0.2605	0.0025	0.5890

Panel B: Reduced Sample (2012-2013 and 2016-2017)

VARIABLES	(1) N	(2) mean	(3) sd	(4) p10	(5) p25	(6) p50	(7) p75	(8) p90	(9) min	(10) max
Loan Lease Yield	4,172	0.0492	0.0160	0.0381	0.0420	0.0466	0.0522	0.0599	0.0184	0.136
Provision Ratio	4,172	0.00361	0.00681	-0.000356	0.000502	0.00192	0.00422	0.00930	-0.00993	0.0399
Z-Score	4,987	3.688	0.631	3.063	3.388	3.638	3.932	4.339	2.075	6.284
Z-Score (CE)	4,628	3.676	0.584	3.103	3.389	3.630	3.907	4.296	2.138	6.105
Leverage	4,643	7.346	3.078	1.459	6.352	7.834	9.078	10.44	0.00762	14.22
Parent Debt Ratio	4,991	0.0199	0.0340	0	0	0.00815	0.0245	0.0608	0	0.336
TobinQ	5,027	1.032	0.0538	0.979	0.998	1.024	1.056	1.089	0.927	1.502
Ln TA	5,026	14.78	2.315	12.28	13.72	14.68	16.04	17.62	0.693	21.49
HQLA Ratio	4,991	0.151	0.0776	0.0653	0.0929	0.138	0.196	0.256	0.00223	0.589

Panel C: Large Non-GSIB BHCs during the Pre-and Post-Period (2012-2013 and 2016-2017)

	2012-2103				2016-2017		
VARIABLES	N	mean	sd	p50	N	mean	sd	p50
HQLA Ratio	36	0.155	0.0440	0.156	36	0.206	0.0472	0.211
Z-Score	34	3.676	0.750	3.638	34	3.536	0.881	3.569
Z-Score (CE)	26	3.666	0.808	3.653	26	3.606	0.722	3.591
Provision Ratio	20	0.00801	0.0104	0.00324	20	0.00951	0.0150	0.00305
Loan Lease Yield	20	0.0543	0.0333	0.0435	20	0.0533	0.0359	0.0403
Leverage	26	4.984	3.334	6.386	26	5.491	3.559	6.788
TobinQ	35	1.048	0.0913	1.014	35	1.061	0.0700	1.042
Ln TA	36	15.15	3.862	16.17	36	15.33	4.113	16.98
Parent Debt Ratio	36	0.0297	0.0341	0.0279	36	0.0285	0.0227	0.0277

Table 3

Parent debt ratio and subsidiary bank's risk-taking. This table presents the results from the model for our preliminary analysis used in Section (4). The dependent variables are the Loan Lease Yield, Provision Ratio, the logarithm of Z-score (Z-Score), the logarithm of Z-score (CE) (Z-Score (CE)), and Leverage. The key explanatory variable is parent debt ratio. For each risk-taking measure, Models (1), (3), (5), (7) and (9) contain only parent debt ratio as a control, while Models (2), (4), (6), (8) and (10) contain all set of controls. All models use quarterly data for 2011-2021Q2. All models include both subsidiary bank and quarter fixed effects. t-statistics reported in the paratheses are based on robust standard error. Superscripts ***, **, * indicates statistical significance at 1%, 5%, 10% level.

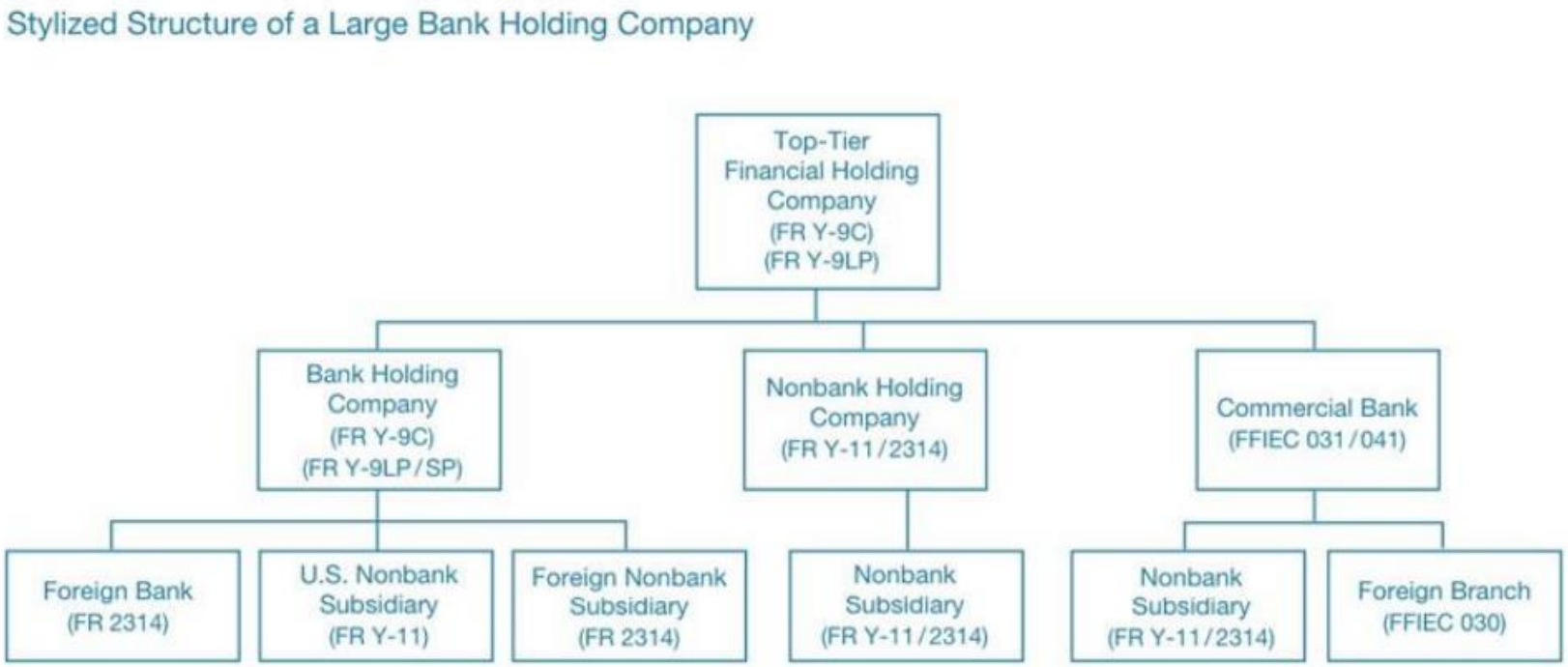
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
VARIABLES	Loan Lease Yield	Loan Lease Yield	Provision Ratio	Provision Ratio	Z-Score	Z-Score	Z-Score (CE)	Z-Score (CE)	Leverage	Leverage
Parent Debt Ratio	-0.068**	-0.062*	-0.047***	-0.048***	3.966***	2.967***	3.951***	2.973***	-17.295***	-16.905***
	(0.035)	(0.037)	(0.017)	(0.018)	(1.364)	(1.132)	(1.335)	(1.122)	(4.655)	(4.413)
Ln TA		-0.001		0.000		-0.156*		-0.153*		0.041
		(0.001)		(0.000)		(0.092)		(0.089)		(0.272)
Leverage		0.000		0.000		-0.054***		-0.053***		
		(0.000)		(0.000)		(0.013)		(0.012)		
HQLA Ratio		-0.001		-0.008**		0.188		0.147		4.314***
		(0.007)		(0.003)		(0.243)		(0.244)		(0.957)
TobinQ		0.013		-0.004		-0.352		-0.400		-0.392
		(0.010)		(0.004)		(0.336)		(0.336)		(1.190)
Constant	0.049***	0.052***	0.004***	0.002	3.579***	6.662***	3.582***	6.659***	7.608***	6.744
	(0.001)	(0.017)	(0.000)	(0.008)	(0.025)	(1.611)	(0.025)	(1.556)	(0.086)	(4.492)
Observations	8,907	8,907	8,907	8,907	9,871	9,871	9,871	9,871	9,898	9,898
R-squared	0.869	0.870	0.571	0.573	0.655	0.671	0.653	0.669	0.878	0.881
Bank Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarter Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Table 4

Difference-in-Differences Analysis in Subsidiary Bank's Risk-taking. This table reports results from difference-in-differences regressions using Equation (2). The treated group includes BHCs with total consolidated assets equal to or greater than 50 billion dollars as of 2014Q1 but excludes GSIBs, while the control group consists of BHCs with total consolidated assets less than 50 billion dollars as of 2014Q1. Post is an indicator variable equal to zero before 2014 and one after 2015. All columns collapse each pre- and post-period into a single observation for each subsidiary bank. Models (1), (3), (5), (7) and (9) contain Nonbank and Time as controls, while Models (2), (4), (6), (8) and (10) contain all set of controls. All models include subsidiary bank fixed effect. T-statistics reported in the paratheses are based on robust standard error. Superscripts ***, **, * indicates statistical significance at 1%, 5%, 10% level.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
VARIABLES	Loan Lease Yield	Loan Lease Yield	Provision Ratio	Provision Ratio	Z-Score	Z-Score	Z-Score (CE)	Z-Score (CE3.65)	Leverage	Leverage
Post	-0.007***	-0.008***	-0.002***	-0.002***	0.096***	0.139*	0.098***	0.254***	-0.347***	-0.399**
	(0.000)	(0.001)	(0.000)	(0.001)	(0.031)	(0.080)	(0.031)	(0.072)	(0.075)	(0.168)
TreatBHC x Post	0.006***	0.006***	0.003***	0.003**	-0.225*	-0.244**	-0.139**	-0.236***	0.854***	0.791***
	(0.002)	(0.002)	(0.001)	(0.001)	(0.121)	(0.122)	(0.065)	(0.086)	(0.234)	(0.261)
GSIBs x Post	-0.004	-0.004	-0.001	0.001	-0.138	-0.108	-0.100	-0.181	0.230	0.143
	(0.005)	(0.005)	(0.002)	(0.002)	(0.128)	(0.126)	(0.121)	(0.134)	(0.304)	(0.287)
TobinQ		0.016		-0.029***		-3.013**		-3.243***		2.989
		(0.019)		(0.009)		(1.256)		(1.241)		(2.870)
Ln TA		0.001		0.002***		0.128		-0.108		0.004
		(0.002)		(0.001)		(0.124)		(0.093)		(0.272)
Leverage		-0.000		0.001***						
		(0.001)		(0.000)						
HQLA Ratio		0.008		-0.021***		-0.196		0.050		1.809
		(0.012)		(0.007)		(0.604)		(0.603)		(1.779)
Constant	0.053***	0.022	0.004***	-0.003	3.606***	4.846**	3.602***	8.475***	7.782***	4.395
	(0.000)	(0.046)	(0.000)	(0.017)	(0.015)	(2.298)	(0.014)	(1.998)	(0.035)	(5.005)
Observations	634	634	634	634	728	728	682	682	690	690
R-squared	0.936	0.937	0.792	0.817	0.741	0.750	0.764	0.772	0.947	0.947
Bank Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Figure 1: The organizational structure of a large BHC



Data sources: Avraham et al. (2012)

Figure 2: Parent debt ratio over time

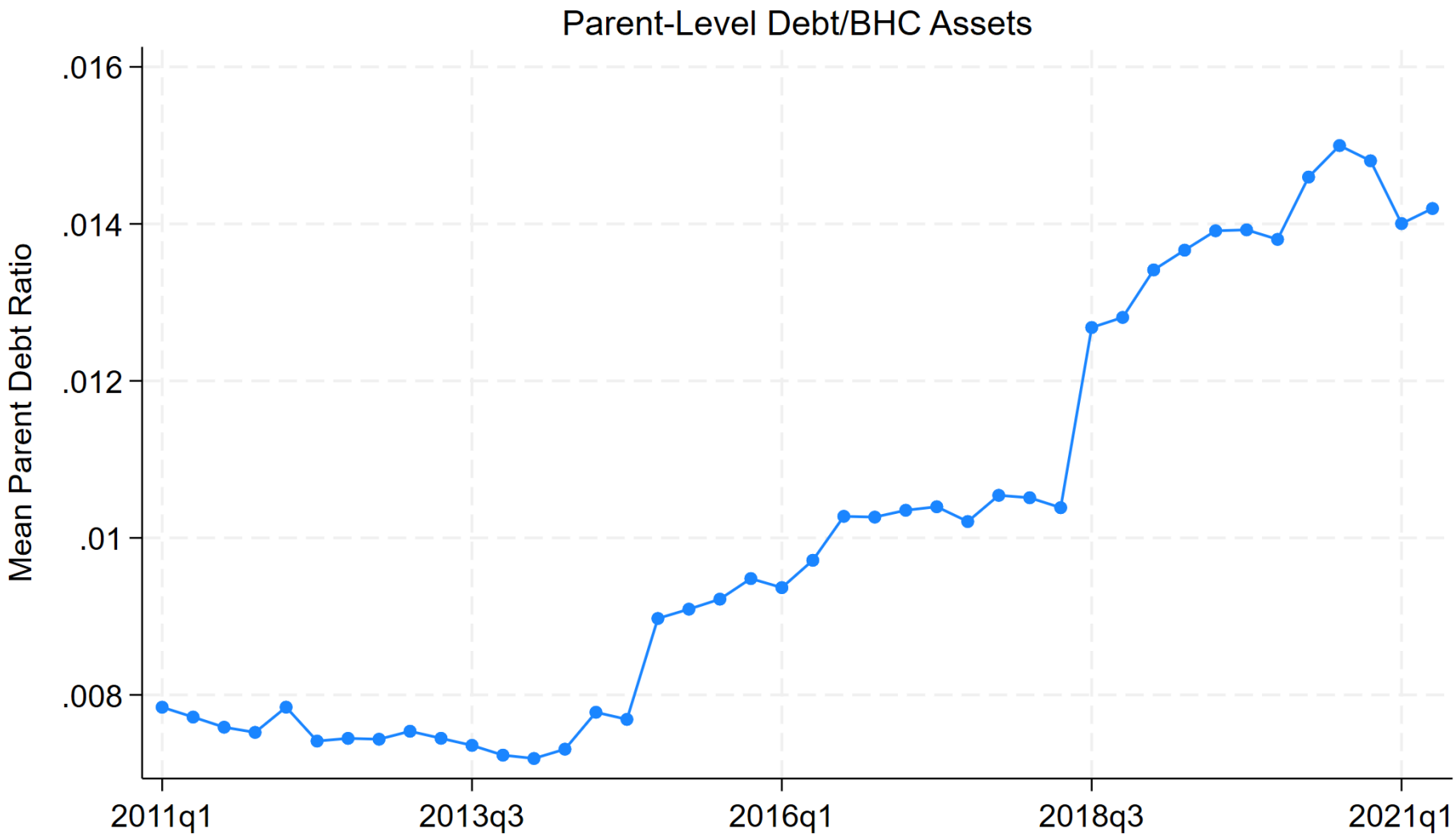


Figure 3: Subsidiary bank's asset ratio over time

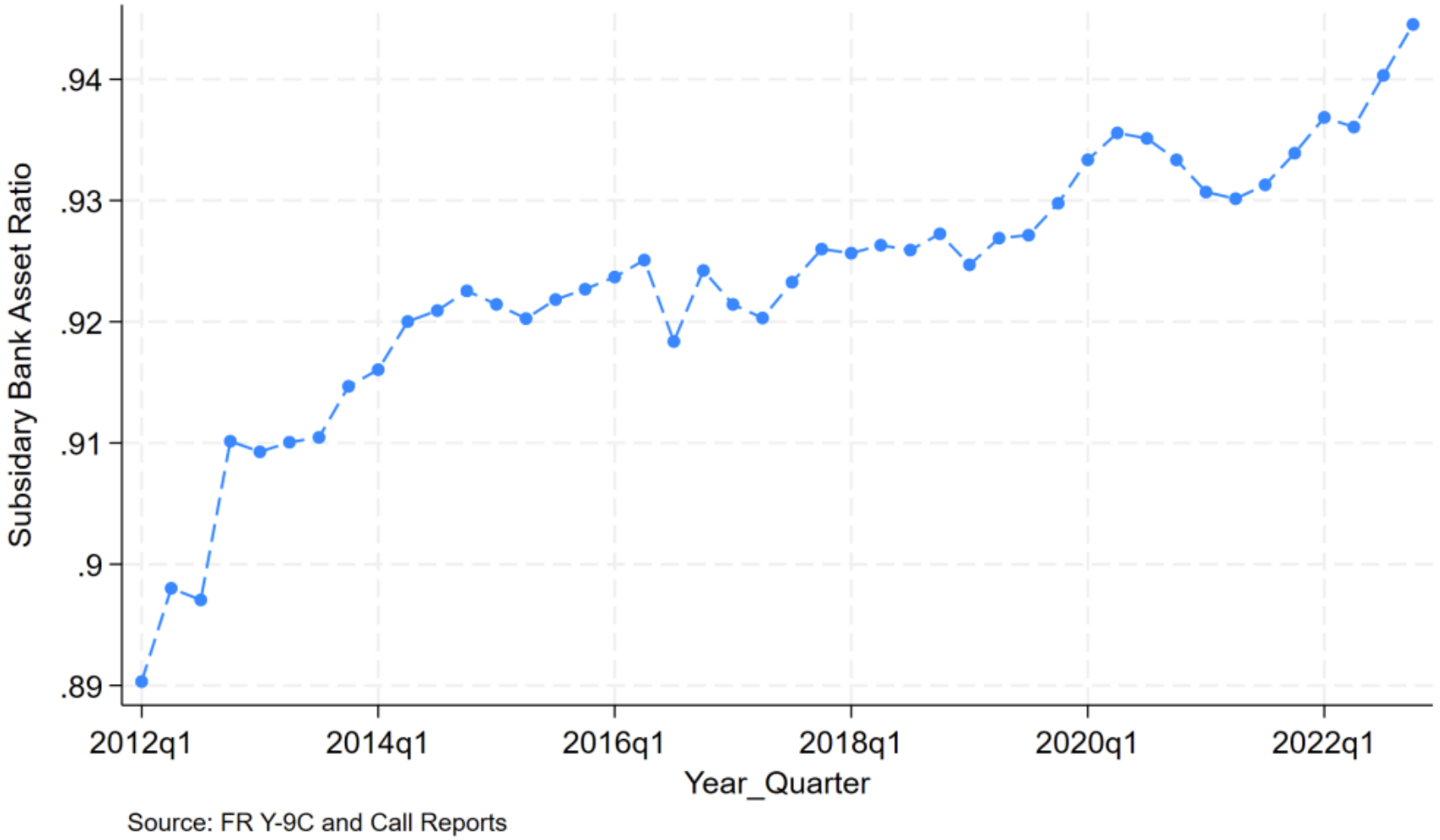


Figure 4: Parallel trend for large non-GSIB BHCs and the control group

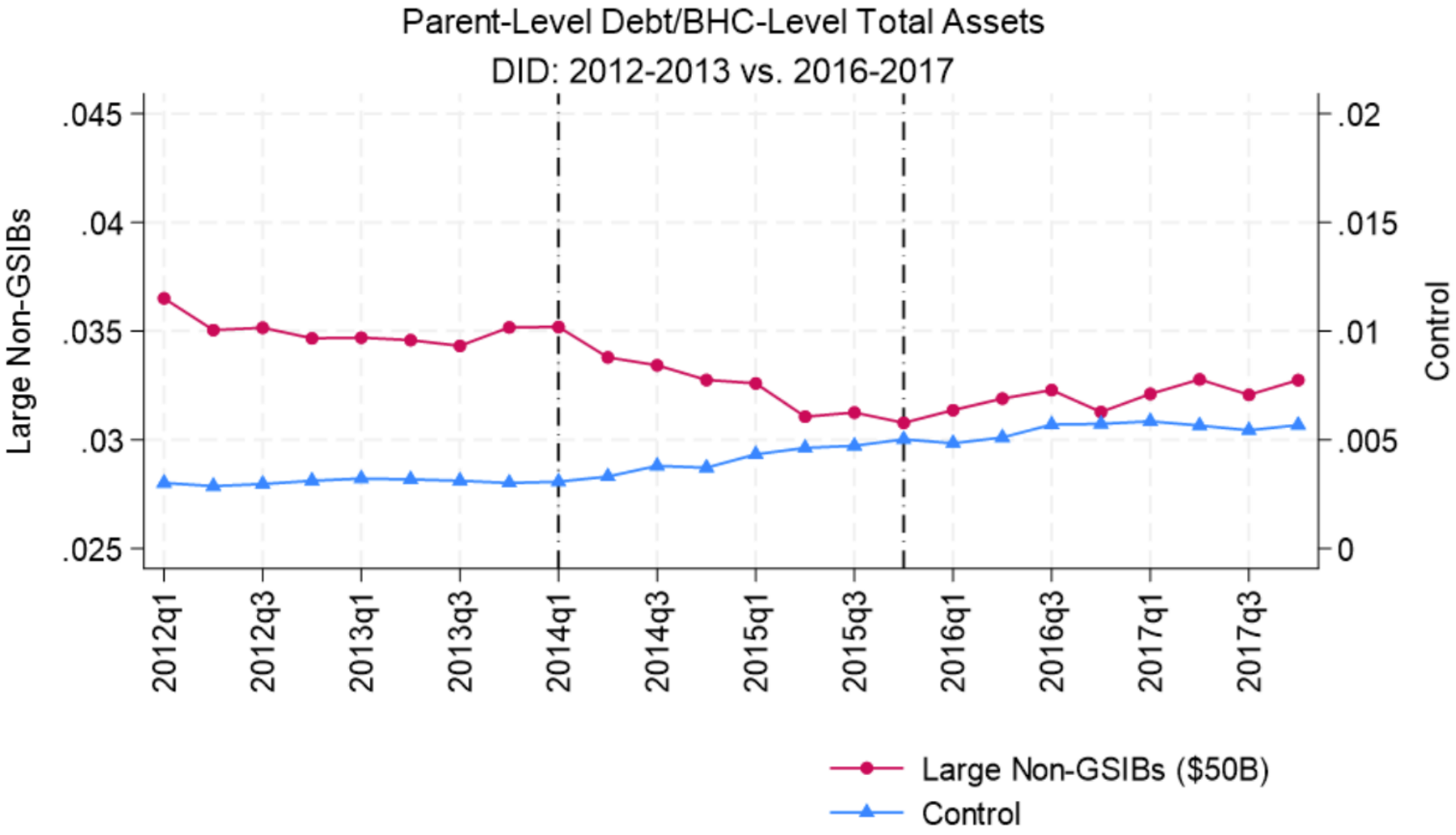


Figure 5: Parallel trend for GSIBs and the control group

